Hamromaster

COMPLETE NOTES

For Hamromaster Online Classes

8.2.
10.29

Correlation:

The relationship between two variables such that change in the value of one variable makes a change in the value of other variable is known as correlation.

Methods of studying correlation:

1) Graphic method (or scatter diagram)
2) Karl Pearson's correlation coefficient
3) Spearman's Rank coefficient
4) Kendall Tau correlation.

Karl Pearson's correlation coefficient.

Formula:- correlation coefficient $(r) = \dfrac{\mathrm{cov}(x,y)}{\sqrt{\mathrm{var}(x)} \cdot \sqrt{\mathrm{var}(y)}}$

where, $\mathrm{cov}(x,y) = \frac{1}{n}(x-\bar{x})(Y-\bar{Y})$

$\mathrm{var}(x) = \sqrt{\frac{1}{n}\Sigma(x-\bar{x})^2}$

$\mathrm{var}(Y) = \sqrt{\frac{1}{n}\Sigma(Y-\bar{Y})^2}$

i) Direct method correlation coefficient

$$r = \frac{n\Sigma XY - \Sigma X \Sigma Y}{\sqrt{n\Sigma X^2-(\Sigma X)^2} \cdot \sqrt{n\Sigma Y^2-(\Sigma Y)^2}}$$

ii) Assumed mean method (or coding method)

$$r = \frac{n\Sigma uv - \Sigma u \, \Sigma v}{\sqrt{n\Sigma u^2-(\Sigma u)^2} \cdot \sqrt{n\Sigma v^2-(\Sigma v)^2}}$$

where, $u = X-A$ (assumed value)

$v = Y-B$ (assumed value)

iii) Actual mean method

$$r = \frac{n\Sigma xy}{\sqrt{\Sigma x^2} \cdot \sqrt{\Sigma y^2}}$$

where, $x = X-\bar{X}$

$y = y-\bar{y}$

correlation lies between

$-1 \le r \le +1$.

8) Pg. 143

Calculate Karl Pearson's coefficient of correlation using:
a) direct method
b) deviation from assumed mean method for the following ages of husband & wives at the time of marriage.

Age of husband (x)	Ages of wives (y)	x^2	y^2	xy
23	18	529	324	414
27	20	729	400	540
28	22	784	484	616
28	27	784	~~736~~ 729	756
28	21	784	441	588
30	29	900	841	870
30	27	900	729	810
~~35~~ 33	29	1089	881	957
35	28	1225	784	980
38	29	1444	881	1102
$\Sigma x = 330$	$\Sigma y = 250$	$\Sigma x^2 = 9168$	$\Sigma y^2 = 6414$	7633

$n = 10$

Result:

$\Sigma x = 330$

$\Sigma y = 250$

$\Sigma x^2 = 9168$

$\Sigma y^2 = 6414$

$\Sigma xy = 7633$

Correlation coefficient $(r) = \dfrac{n\Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2} \cdot \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$

$= \dfrac{10 \times 7633 - 330 \times 250}{\sqrt{10 \times 9168 - (330)^2} \cdot \sqrt{10 \times 6414 - (250)^2}}$

$= 0.8013$ #

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Rank correlation

Case (II) when ranks are repeated: If two or more variate values are equal then each equal value is given the common rank. In such cases each individual is given an average rank.

Formula: $R = 1 - \dfrac{6\left\{\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \dots\right\}}{n(n^2-1)}$

palpali bharat

Eg:-

Marks in account	Marks in stats	Rank (R_1)	Rank (R_2)	$d=R_1-R_2$	d^2
15	40	7	3	4	16
20	30	5.5	5	0.5	0.25
28	50	4	2	2	4
12	30	8	5	3	9
40	20	9	7	2	4
60	10	1	8	-7	49
20	30	5.5	5	0.5	0.25
80	60	1	1	0	0
					$\sum d^2 = 81.5$

The value 20 is repeated 2 times!

$\therefore m_1 = 2$

The value 30 is repeated 3 times

$\therefore m_2 = 3$

$$R = 1 - \frac{6\left\{\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12}\right\}}{n(n^2-1)}$$

$$= 1 - \frac{6\left\{81.5 + \frac{2(2^2-1)}{12} + \frac{3(3^2-1)}{12}\right\}}{8(8^2-1)}$$

$$= 1 - 6 \quad 0$$

Hence, there is no correlation betn both subject.

Example : 2 Pg : 143

Calculate Rank correlation from the given data.

	Age of husband	Age of wife	Rank (R_1)	Rank (R_2)	$d = R_1 - R_2$	d^2
	23	18	10	10	0	0
	27	20	9	9	0	0
m_2	28	22	7	7	0	0
m_2	28	27	7	5.5	2.5	6.25
m_2	28	21	7	8	-1	1
m_1	30	29	4.5	2	2.5	6.25
m_1	30	27	4.5	5.5	1.5	2.25
	33	29	3	2	1	1
	35	28	2	4	2	4
	38	29	1	2	-1	1
						21.75

R = highest = 38
for 38 = 1

For 30 → $\frac{4+5}{2} = 4.5$

28 → $\frac{6+7+8}{3} = 7$

30 → 4
→ 5
28 → 6
28 → 7
28 → 8

For, 29
$\frac{1+2+3}{3} = 2$

$\frac{5+6}{2}$

Here, the value 30 is repeated 2 times. ∴ $m_1 = 2$
the value 28 " " 3 " ∴ $m_2 = 3$
Again, the value 29 is repeated 3 times ∴ $m_3 = 3$
" " 27 " " 2 " ∴ $m_4 = 2$.

$$\therefore R = 1 - \frac{6\left\{\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12} + \frac{m_4(m_4^2-1)}{12}\right\}}{n(n^2-1)}$$

$$= 1 - 6\left\{21.75 + 2\frac{(4-1)}{12} + 3\frac{(9-1)}{12} + 3\frac{(9-1)}{12} + 2\frac{(4-1)}{12}\right\}$$

Kendall Tau Rank Correlation.

The degree of relationship between ranks of two variable x and y can also be establish by the method given by M. G. Kendell.

Formula : $t = \frac{2S}{n(n-1)}$

where, t = Kendell Tau Rank Correlation.
S = sum of scores.
n = number of pairs.

Process of calculations:

Step 1: Rank the given variables x & y

Step 2: Arrange the ranks of x in natural sequence & ranks of y corresponding.

Step 3: Take the first rank of y & give +1 mark or -1 marks value for all pairs with subsequent rank of y if it is greater than given +1 & it is less than given mark -1. The sum of these marks considered as the scores of the particular rank.

Step 4: Similarly obtain the score of other ranks also till we reach the last pair.

Step 5: Find the sum of all these scores.

In case of tied observations (repeated rank) Kendell tau rank correlation is given by:

$$\tau = \frac{S}{\sqrt{\frac{1}{2}n(n-1) - \Sigma T_x}\sqrt{\frac{1}{2}n(n-1) - \Sigma T_y}}$$

where, $T_x = \frac{t_x(t_x - 1)}{2}$

$T_y = \frac{t_y(t_y - 1)}{2}$

Eg:

Following are the marks according to six dancers in a dance competition by two judge.

Dancers	Marks by Judge A	Marks by Judge B	R_1	R_2
B	15	20	3	4
C	18	16	4	1
D	10	30	1	6
E	20	18	5	3
F	25	25	6	5
G	14	17	2	2

Step 3: We write below ranks of x in natural sequence and ranks of y corresponding.

rank of x	1	2	3	4	5	6
rank of y	6	2	4	1	3	5

To calculate δ take the rank

Dancers	Marks by Judge A	Marks by Judge B	R_1	R_2
B	15	20	3	4
C	18	16	4	1
D	10	30	1	6
E	20	18	5	3
F	25	25	6	5
G	14	17	2	2

Step 3: We write below ranks of x in natural sequence and ranks of y corresponding.

ranks of x	1	2	3	4	5	6
ranks of y	6	2	4	1	3	5

To calculate the value of S

Take the rank 6

(6,2) , (6,4) , (6,1) , (6,3) , (6,5)

Take the rank 2

(2,4) , (2,1) , (2,3) , (2,5)

Take the rank 4

(4,1) , (4,3) , (4,5)

Take the rank 1

(1,3) , (1,5)

Take the rank 3

(3,5)

S = score of rank 6 + score of rank 2 + score of rank 4 - score of rank 1 + score of rank 3

$= (-1-1-1-1-1) + (+1-1+1+1) + (-1+1+1) + (+1+1) + 1$

$= -5 + 2 - 1 + 2 + 1$

$= -1.$

$$\tau = \frac{2S}{n(n-1)}$$

$$= \frac{2 \times -1}{6(6-1)}$$

$$= \frac{-2}{6 \times 5}$$

$$= -\frac{1}{15}$$

$= -0.067.$

(V.V.I)

Interpretation of correlation coefficient

(i) If $r = +1$, there is perfect positive correlation between two variables.

(ii) If $r = -1$, there is perfect Negative correlation between two variables.

(iii) If $r = 0$, there is no correlation bet^n two variables.

(iv) If r lies between 0.001 to 0.499 there is low degree of positive correlation.

(v) If r lies between 0.5 to 0.699 there is moderate positive correlation.

(vi) If r lies between 0.70 to 0.999 High degree positive correlation.

(vii) If r lies between -0.499 to -0.001, there is Low degree of Negative Correlation.

(viii) If r lies between -0.699 to -0.500 there is moderate Negative correlation.

(ix) If r lies between -0.699 to -0.999 there is high degree of Negative correlation.

• Karl pearson's coefficient of corr[illegible]n